Ideal Fluid Flow in a Polygonal Domain via the Schwarz–Christoffel Transformation

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Abstract

This paper applies the Schwarz–Christoffel (SC) conformal mapping to simulate steady, irrotational, incompressible ideal fluid flow inside an 11-vertex simplification of the Boulder, Colorado city-boundary polygon. The SC transformation maps the upper half-plane $\mathbb{H}$ conformally onto the polygonal domain Ω, converting analytically tractable complex potentials in $\mathbb{H}$ into streamline and equipotential patterns in the physical domain. Two flow models are derived and implemented: a uniform ideal flow $W = U\zeta$ providing the baseline conformal grid, and a doubly-connected flow in which a downtown obstacle is modelled via the Milne-Thomson circle theorem. The SC parameter problem is solved by softmax pre-vertex parameterisation combined with Levenberg–Marquardt least squares on side-length ratios computed by Gauss–Legendre quadrature. All results are produced through the forward SC map, yielding streamlines that satisfy the no-penetration boundary condition exactly by construction. A closed-form verification on the rectangle-to-half-plane map confirms the implementation.

1 Introduction

Computing fluid flow in an arbitrary domain is a hard problem: Laplace's equation $\nabla^2\phi = 0$ admits no elementary closed-form solution in an irregular geometry. The classical approach uses conformal mapping. A conformal map $z = f(\zeta)$ from a canonical domain (the unit disk or the upper half-plane) to the physical region Ω transforms the Laplace equation from one domain to the other while preserving harmonicity, so that any analytic function $W(\zeta)$ satisfying the required boundary conditions in the canonical domain produces a valid solution in Ω via $\tilde{W}(z) = W(f^{-1}(z))$.

The Schwarz–Christoffel (SC) transformation, developed independently by H. A. Schwarz and E. B. Christoffel in the 1860s, gives an explicit integral formula for the conformal map from the upper half-plane $\mathbb{H}$ to any simply connected polygon. For analytic polygons the parameter problem (choosing pre-images of the vertices on the real axis) can sometimes be solved in closed form; for general n-gons it must be treated numerically. The modern numerical theory was systematised by Driscoll and Trefethen [Driscoll and Trefethen, 2002]; the classical theory is in Nehari [Nehari, 1952] and Ahlfors [Ahlfors, 1979].

We apply the SC transformation to an 11-vertex polygon derived from the Boulder, Colorado city boundary, chosen for its mix of convex and reflex vertices as a non-trivial test of the numerical pipeline. Two flow regimes are considered: (i) uniform ideal flow, providing the baseline conformal grid, and (ii) doubly-connected flow with an interior obstacle modelled via the Milne-Thomson circle theorem. The polygon's geometric complexity exercises the full pipeline; any physical interpretation as "airflow over a city" is illustrative. The mathematical framework is drawn from Ablowitz and Fokas [2003], Driscoll and Trefethen [2002], Nehari [1952]; the fluid-dynamics background from Bender and Orszag [1999] and Milne-Thomson [1968].

A closed-form verification against the rectangle-to-half-plane map (an elliptic-integral identity) is included to certify the numerical pipeline independently of the Boulder application.

Organisation. Section 2 develops the SC mapping theory and complex-potential framework, with step-by-step derivations of every key result. Section 3 describes the numerical algorithms and verifies the pipeline against the closed-form rectangle map. Section 4 applies the method to the Boulder polygon and presents results for both flow models. Section 5 summarises the conclusions. Technical details on the Levenberg–Marquardt solver are in Appendix A; code availability in Appendix B.

2 Mathematical Framework

2.1 Ideal Fluid Flow and the Complex Potential

A steady two-dimensional flow is *ideal* when the velocity field $\mathbf{u} = (u, v)$ is simultaneously irrotational ($\partial_x v - \partial_y u = 0$) and incompressible ($\partial_x u + \partial_y v = 0$). Irrotationality in a simply connected region forces $\mathbf{u}$ to be a gradient field $\mathbf{u} = \nabla\phi$ (velocity potential), and incompressibility then gives $\nabla^2\phi = 0$. Incompressibility independently forces $\mathbf{u}$ to be the skew-gradient of a stream function ψ via $u = \partial_y\psi$, $v = -\partial_x\psi$, and irrotationality then gives $\nabla^2\psi = 0$. Comparing the two representations produces the Cauchy–Riemann equations

$$\partial_x\phi = \partial_y\psi, \qquad \partial_y\phi = -\partial_x\psi, \tag{1}$$

so the *complex potential*

$$W(z) = \phi(x, y) + i\,\psi(x, y), \qquad z = x + iy, \tag{2}$$

is analytic on the flow domain.

Velocity from $W'(z)$. For any analytic function the complex derivative can be evaluated along the real direction: $W'(z) = \partial_x W = \partial_x \phi + i\,\partial_x \psi$. Using $\partial_x \phi = u$ and the Cauchy–Riemann equation $\partial_x \psi = -\partial_y \phi = -v$, we obtain

$$W'(z) = u - iv = \overline{u + iv}, \qquad \mathbf{u} = \overline{W'(z)}. \tag{3}$$

Physical interpretation. Level curves of ϕ are *equipotentials*; level curves of ψ are *streamlines*. The Cauchy–Riemann equations imply $\nabla\phi \cdot \nabla\psi = 0$, so streamlines and equipotentials meet orthogonally wherever $W'(z) \neq 0$. The no-penetration condition on a solid boundary $\partial\Omega$ reads $\mathbf{u} \cdot \hat{\mathbf{n}} = 0$, equivalent to $\psi = \text{const}$ along $\partial\Omega$.

2.2 Conformal Invariance of Harmonicity

Lemma 1. *Let $f : \mathbb{H} \to \Omega$ be a bijective analytic map with $f'(\zeta) \neq 0$. If $W(\zeta)$ is analytic on $\mathbb{H}$, then $\tilde{W}(z) = W(f^{-1}(z))$ is analytic on Ω; in particular, both $\tilde{\phi} = \mathrm{Re}\tilde{W}$ and $\tilde{\psi} = \mathrm{Im}\tilde{W}$ satisfy Laplace's equation on Ω.*

This is the structural fact that makes conformal mapping work for ideal flow. Any solution of Laplace's equation in $\mathbb{H}$ pulls back to a solution in Ω, and the no-penetration condition $\psi = 0$ on $\partial\mathbb{H}$ (easy to arrange by the method of images) becomes $\psi = 0$ on $\partial\Omega$ automatically.

2.3 The Riemann Mapping Theorem

Theorem 2 (Riemann, 1851)**.** *Let $\Omega \subsetneq \mathbb{C}$ be a simply connected domain. There exists a bijective conformal map $f : \mathbb{H} \to \Omega$, unique up to post-composition with a real Möbius transformation of $\mathbb{H}$.*

The Boulder polygon is simply connected, so f exists. The three real parameters of the Möbius freedom correspond to fixing three real pre-images on $\partial\mathbb{H} = \mathbb{R} \cup \{\infty\}$.

2.4 The Schwarz–Christoffel Formula

For a polygon Ω with vertices $w_1, \ldots, w_n$ (counter-clockwise) and interior angles $\alpha_k\pi$, $\alpha_k \in (0, 2)$, the SC map takes the form [Nehari, 1952, Ch. 5][Driscoll and Trefethen, 2002, Ch. 5]

$$f(\zeta) = A + C \int_{\zeta_0}^{\zeta} \prod_{k=1}^{n} (t - \zeta_k)^{\alpha_k - 1}\, \mathrm{d}t, \tag{4}$$

where $\zeta_k \in \mathbb{R}$ are the *pre-vertices*, $A \in \mathbb{C}$ is a translation, and $C \in \mathbb{C}$ encodes scale and rotation. We derive the four structural properties that force (4).

Step 1: Polygon angle-sum identity. For a simple CCW polygon the exterior turning angle at vertex k is $\pi - \alpha_k\pi = \pi(1 - \alpha_k)$. The total exterior turning over a full CCW traversal equals 2π, hence

$$\sum_{k=1}^{n} \pi(1 - \alpha_k) = 2\pi \quad \Longleftrightarrow \quad \sum_{k=1}^{n} \alpha_k = n - 2. \tag{5}$$

Step 2: Direction of $f'(\zeta)$ on the real axis. Take $\zeta = t$ real with $t \in (\zeta_k, \zeta_{k+1})$. For each factor $(t - \zeta_j)^{\alpha_j - 1}$:

- if $j \le k$ (so $t > \zeta_j$), the base is positive real and $\arg[(t - \zeta_j)^{\alpha_j - 1}] = 0$;
- if $j > k$ (so $t < \zeta_j$), the base is negative real and $\arg[(t - \zeta_j)^{\alpha_j - 1}] = (\alpha_j - 1)\pi$ (principal branch, $\arg(-1) = \pi$).

Therefore

$$\arg f'(t) = \arg C + \pi \sum_{j=k+1}^{n} (\alpha_j - 1). \tag{6}$$

This is constant on the open interval (ζ_k, ζ_{k+1}), so the image $f\big((\zeta_k, \zeta_{k+1})\big)$ is a straight line segment, which is the k-th polygon edge.

As t increases past ζ_k, the contribution of the k-th factor to $\arg f'$ decreases by $(\alpha_k - 1)\pi$, i.e., the tangent direction turns CCW by $(1 - \alpha_k)\pi$. This is exactly the exterior turning angle at w_k, consistent with an interior angle of $\alpha_k \pi$.

Step 3: Vertex angle reproduction. The direction jump at vertex w_k is the exterior turning $(1 - \alpha_k)\pi$, so the interior angle is $\pi - (1 - \alpha_k)\pi = \alpha_k \pi$, reproducing the prescribed geometry.

Step 4: Convergence at infinity. If $\zeta_n = \infty$ (one vertex is sent to infinity) the factor $(t - \zeta_n)^{\alpha_n - 1}$ is dropped. Near $t \to \infty$ the remaining integrand behaves as $t^{\sum_{j=1}^{n-1} (\alpha_j - 1)}$. Using (5),

$$\sum_{j=1}^{n-1} (\alpha_j - 1) = \left(\sum_{j=1}^{n} \alpha_j \right) - \alpha_n - (n-1) = (n-2) - \alpha_n - (n-1) = -\alpha_n - 1, \tag{7}$$

so the integrand decays as $t^{-\alpha_n - 1}$ and the integral converges at infinity provided $\alpha_n > 0$.

Interior-angle formula in practice. Given vertices $w_0, \ldots, w_{n-1}$ (CCW), the signed exterior turning at w_k is $\arg\big((w_{k+1} - w_k)/(w_k - w_{k-1})\big)$, so the interior angle in units of π is

$$\alpha_k = 1 - \frac{1}{\pi} \arg\left(\frac{w_{k+1} - w_k}{w_k - w_{k-1}} \right), \tag{8}$$

with indices modulo n.

2.5 Möbius Normalisation and the Parameter Problem

The real Möbius group $\mathrm{PSL}(2, \mathbb{R})$, acting on $\mathbb{H}$ via $\zeta \mapsto (a\zeta + b)/(c\zeta + d)$, has three real parameters. Three of the n pre-vertices can therefore be fixed at prescribed values without loss of generality. We use

$$\zeta_0 = -1, \qquad \zeta_1 = 0, \qquad \zeta_{n-1} = 1, \tag{9}$$

leaving $n - 3$ free parameters $\zeta_2 < \cdots < \zeta_{n-2}$ in $(0, 1)$. For Boulder, $n = 11$ yields $n - 3 = 8$ free pre-vertices. These are determined by matching side-length ratios; see Section 3.2.

2.6 Method of Images

Any analytic $W(\zeta)$ with $\mathrm{Im}W = 0$ on $\mathbb{R}$ automatically satisfies $\psi = 0$ on $\partial\Omega$ after pull-back via f. To add a singularity at $s = a + ib \in \mathbb{H}$ (with $b > 0$) while preserving $\mathrm{Im}W = 0$ on the real axis, we pair it with a mirror-image singularity at $\bar{s} = a - ib$. For a source of strength $Q > 0$ the image-pair potential is

$$W_{\mathrm{source}}(\zeta) = \frac{Q}{2\pi}\big[\log(\zeta - s) + \log(\zeta - \bar{s})\big]. \tag{10}$$

Algebraic verification. Take $\zeta = x \in \mathbb{R}$:

$$\log(x - s) + \log(x - \bar{s}) = \log\!\big[(x-a)^2 + b^2\big] + i\,\big[\arg(x - a - ib) + \arg(x - a + ib)\big]. \tag{11}$$

The numbers $x - a - ib$ and $x - a + ib$ are complex conjugates, so their arguments are opposite in sign, making the bracketed imaginary part vanish. Therefore

$$\mathrm{Im}\big[\log(x - s) + \log(x - \bar{s})\big] = 0 \quad \text{for all } x \in \mathbb{R}. \tag{12}$$

2.7 Milne-Thomson Circle Theorem

Theorem 3 (Milne-Thomson, 1940)**.** *Let $W_0(\zeta)$ be analytic on $\mathbb{C} \setminus D$ where $D = \{\zeta : |\zeta - \zeta_0| \le a\}$, with no singularities on ∂D. Then*

$$W^{\mathrm{ext}}(\zeta) = W_0(\zeta) + \overline{W_0\!\left(\zeta_0 + \frac{a^2}{\overline{\zeta - \zeta_0}}\right)} \tag{13}$$

is analytic on the exterior of D, has the same singularity structure there as W_0, and satisfies $\mathrm{Im}W^{\mathrm{ext}} = 0$ on ∂D.

Proof. Streamline on ∂D. Write $\zeta - \zeta_0 = ae^{i\theta}$ so that $|\zeta - \zeta_0| = a$ and

$$(\zeta - \zeta_0)\,\overline{(\zeta - \zeta_0)} = a^2 \quad\Longrightarrow\quad \frac{a^2}{\overline{\zeta - \zeta_0}} = \zeta - \zeta_0, \tag{14}$$

hence on ∂D: $\zeta_0 + a^2/\overline{\zeta - \zeta_0} = \zeta_0 + (\zeta - \zeta_0) = \zeta$. Substituting into (13) gives

$$W^{\mathrm{ext}}(\zeta)\big|_{\partial D} = W_0(\zeta) + \overline{W_0(\zeta)} = 2\,\mathrm{Re}W_0(\zeta) \in \mathbb{R}, \tag{15}$$

so $\mathrm{Im}W^{\mathrm{ext}} = 0$ on ∂D.

Analyticity outside D via Schwarz reflection. Define $\iota(\zeta) = \zeta_0 + a^2/\overline{\zeta - \zeta_0}$, which maps $\mathbb{C} \setminus \overline{D}$ bijectively onto $D \setminus \{\zeta_0\}$. Since ι depends on $\overline{\zeta}$ rather than ζ, the composition $W_0 \circ \iota$ is a holomorphic function of $\overline{\zeta}$ on the exterior of D, equivalently an anti-holomorphic function of ζ. Taking the complex conjugate of the entire expression, $\overline{W_0(\iota(\zeta))}$, converts this into a holomorphic function of ζ on $\mathbb{C} \setminus \overline{D}$. This is the Schwarz reflection principle specialised to a circle [Ahlfors, 1979, Ch. 4.6]. Consequently $W^{\mathrm{ext}} = W_0(\zeta) + \overline{W_0(\iota(\zeta))}$ is holomorphic in ζ outside D, and its singularities there are exactly those of W_0 (those of $W_0 \circ \iota$ all lie inside D, where they are excluded by hypothesis from the problem domain). $\square$

Upper-half-plane specialisation. Equation (13) gives a potential with ∂D as a streamline, valid on the whole complex plane exterior to D. To additionally enforce $\psi = 0$ on the real axis $\partial\mathbb{H}$, the method of images from Section 2.6 requires adding a mirror term obtained by reflecting the obstacle centre ζ_0 across the real axis. The mirror image of the dipole at ζ_0 is a dipole of the same strength at $\overline{\zeta_0}$ in the lower half-plane, contributing $Ua^2/(\zeta - \overline{\zeta_0})$. For $W_0 = U\zeta$ the combined expression

$$W_{\text{urban}}(\zeta) = U\zeta + \frac{Ua^2}{\zeta - \zeta_0} + \frac{Ua^2}{\zeta - \overline{\zeta_0}} \tag{16}$$

is holomorphic in ζ (the variable in every denominator is ζ itself), and both conditions hold simultaneously: the first additional term makes ∂D a streamline; the second restores $\psi = 0$ on $\mathbb{R}$.

Approximation error. When the actual obstacle boundary differs from a circle, approximating it by the minimum enclosing circle introduces a leading-order error that scales as $O((a/\operatorname{Im}\zeta_0)^2)$ [Milne-Thomson, 1968, Ch. 6]. This is an order-of-magnitude estimate rather than a rigorous bound. Direct verification against an exact doubly-connected SC map is noted in Section 5 as future work.

2.8 SC Crowding and the Conformal Modulus

A practical obstruction to numerical SC is *crowding*: for polygons with strongly different edge-length scales, the pre-vertices cluster exponentially on the real axis. The phenomenon is not an artefact of the solver; it reflects a rigid complex-analytic invariant.

Conformal modulus of a rectangle. Consider the rectangle $R = [-K, K] \times [0, K']$ of width $2K$ and height K'. Its *conformal modulus* (aspect ratio) is $m(R) = 2K/K'$ [Driscoll and Trefethen, 2002, Ch. 2], and it is a conformal invariant: no conformal map sends R to a rectangle of different modulus. The SC pre-vertices $\{-1/k, -1, 1, 1/k\}$ that map to the corners of R are related to $m(R)$ through the complete elliptic integrals

$$K(k) = \int_0^1 \frac{\mathrm{d}t}{\sqrt{(1-t^2)(1-k^2t^2)}}, \qquad K'(k) = K(\sqrt{1-k^2}), \tag{17}$$

with $m(R) = 2K(k)/K'(k)$ [Olver et al., 2010, §19.2]. As $m(R) \to \infty$ the elliptic modulus $k \to 1$, and the standard asymptotic expansion of $K(k)$ [Olver et al., 2010, §19.12] gives the inner pre-vertex spacing

$$1/k - 1 \;\sim\; 8e^{-\pi m(R)/2} \qquad \text{as } m(R) \to \infty. \tag{18}$$

Consequence for general polygons. For a general polygon containing a long thin region of aspect ratio L, the pre-vertex spacing across that region contracts exponentially in L [Driscoll and Trefethen, 2002, Ch. 2]. For $L = 5$ the inner spacing is already $\approx 10^{-3}$, for $L = 10$ the spacing is $\approx 10^{-7}$, and for $L \gtrsim 13$ it falls below 10^{-9} and becomes difficult to resolve reliably in double precision. This limits SC solvers to $n \lesssim 20$ vertices with mild aspect ratios, and is the reason the 1935-vertex Boulder boundary must be simplified to an 11-vertex polygon (Section 3.1). The chosen simplification has all interior angles confined to $[0.35\pi, 1.75\pi]$, keeping the conformal modulus of each sub-region within the stable regime.

3 Numerical Implementation

3.1 Polygon Preprocessing

The raw Boulder TIGER/Line boundary [US Census Bureau, 2025] has 1935 vertices. Douglas–Peucker simplification with an adaptive binary-search tolerance reduces it to 14 vertices; three vertices with reflex angles exceeding 1.75π are then removed iteratively by merging adjacent edges, yielding the final $n = 11$ polygon. The polygon is normalised to $O(1)$ scale and centred at the origin for numerical stability. The angle bounds $[0.35\pi, 1.75\pi]$ are chosen from the crowding discussion of Section 2.8.

3.2 The SC Parameter Problem

The $n - 3 = 8$ free pre-vertices $\zeta_2, \ldots, \zeta_{n-2} \in (0, 1)$ are chosen so that the image polygon $f(\mathbb{R})$ has the correct side-length ratios. Define the k-th side length

$$L_k = \left| \int_{\zeta_k}^{\zeta_{k+1}} \prod_{j=1}^{n} (t - \zeta_j)^{\alpha_j - 1} \, \mathrm{d}t \right|, \tag{19}$$

and require

$$\frac{L_k}{L_{n-1}} = \frac{|w_{k+1} - w_k|}{|w_0 - w_{n-1}|}, \qquad k = 0, \ldots, n-2. \tag{20}$$

Softmax reparameterisation. Given $p \in \mathbb{R}^{n-3}$, define weights and cumulative pre-vertex positions:

$$w_j(p) = \frac{e^{p_j}}{\sum_{i=1}^{n-3} e^{p_i} + 1}, \qquad \zeta_{k+1} = \sum_{j=1}^{k} w_j(p). \tag{21}$$

Since $e^{p_j} > 0$, each $w_j > 0$ and the cumulative sums are strictly increasing. Let $S = \sum_i e^{p_i}$; then $\sum_j w_j = S/(S+1) < 1$, so every $\zeta_k \in (0, 1)$. The ordering and box constraints are thus automatic, and the optimisation over $p \in \mathbb{R}^{n-3}$ is unconstrained.

Solver and quadrature. System (20) is solved by Levenberg–Marquardt least squares via SciPy's `least_squares` routine. Each side-length integral is evaluated by $N = 500$-node Gauss–Legendre quadrature on $[-1, 1]$ after affine rescaling to $[\zeta_k, \zeta_{k+1}]$; nodes and weights are pre-computed and reused across iterations, so each evaluation is a single vectorised dot product. Levenberg–Marquardt update details are in Appendix A.

Determining A and C. Once the pre-vertices are known, $A, C \in \mathbb{C}$ are obtained from the overdetermined linear system $A + C \cdot I_k = w_k$ for $k = 0, 1, \ldots, n-1$, where I_k is the SC integral from a reference point $\zeta_{\text{ref}} = 0.5i$ to $\zeta_k + \varepsilon$ (with a small complex offset ε to avoid the branch point). The least-squares solution is computed via `numpy.linalg.lstsq`.

3.3 Forward Map and Streamline Computation

All flow results are produced via the *forward-map* approach. A dense 400×400 grid of ζ values in $\mathbb{H}$ is mapped forward to $z = f(\zeta)$, each SC integral evaluated by a 400-node Gauss–Legendre rule along an L-shaped path in $\mathbb{H}$ that clears the real-axis branch cuts. The complex potential $W(\zeta)$ is evaluated analytically at each ζ point. The resulting scattered pairs $(z_i, W(\zeta_i))$ are interpolated onto a regular physical-plane grid (`scipy.interpolate.griddata`, linear method), and contours of $\mathrm{Im} W$ (streamlines) and $\mathrm{Re} W$ (equipotentials) are extracted.

This avoids the per-pixel Newton iteration that inverse mapping would require. Grid points from $\mathbb{H}$ land strictly inside Ω by construction, so the interpolated field is zero outside Ω with smooth contours at the boundary.

3.4 Verification on the Rectangle Map

Before applying the pipeline to Boulder, we verify it against the one polygonal case with a closed-form SC map: the rectangle. The classical construction [Nehari, 1952, Ch. 5] uses the symmetric pre-vertex set $\{-1/k, -1, 1, 1/k\}$ with elliptic modulus $0 < k < 1$, giving

$$f(\zeta) = \int_0^\zeta \frac{\mathrm{d}t}{\sqrt{(1-t^2)(1-k^2t^2)}} = F(\arcsin \zeta,\, k), \tag{22}$$

with F the incomplete elliptic integral of the first kind [Olver et al., 2010, §19.2]. The corner images are $f(\pm 1) = \pm K(k)$ and $f(\pm 1/k) = \pm K(k) + iK'(k)$, so the image rectangle has width $2K(k)$ and height $K'(k)$; its conformal modulus matches Section 2.8.

Adaptation to the project's parameterisation. The softmax reparameterisation (21) constrains all free pre-vertices to $(0, 1)$ and fixes three anchors at $\{-1, 0, 1\}$. Values $\pm 1/k$ with $k < 1$ lie outside this range, so the classical symmetric placement cannot be used directly. The two normalisations are related by the unique real Möbius transformation matching the ordered cross-ratio on the real axis:

$$[-1, 0; s, 1] = [-1/k, -1; 1, 1/k], \tag{23}$$

where $[a, b; c, d] = (a-c)(b-d)/[(a-d)(b-c)]$. Computing both sides,

$$\frac{1+s}{2s} = \frac{(1+k)^2}{4k} \qquad \Longrightarrow \qquad s(1+k)^2 = 2k(1+s) \qquad \Longrightarrow \qquad s = \frac{2k}{1+k^2}. \tag{24}$$

This gives a one-to-one correspondence between $k \in (0, 1)$ in the classical picture and the single free pre-vertex $s \in (0, 1)$ in our pipeline. For a target rectangle of conformal modulus $m(R) = 2K(k)/K'(k)$, the pipeline must recover the corresponding s determined by (24).

Numerical verification. Take $m(R) = 2$ (the standard 2×1 rectangle), which corresponds to $K(k) = K'(k)$, hence $k = 1/\sqrt{2}$ and $s = 2\sqrt{2}/3 \approx 0.9428$. The softmax + Levenberg–Marquardt solver converges to s with relative error below 10^{-8}, limited by the LM convergence tolerance on side-length ratios.

For the SC integrand itself, each sub-integral between adjacent pre-vertices has integrable $(t - \zeta_k)^{-1/2}$ endpoint singularities. We absorb each singularity with the substitution $t - \zeta_k = u^2$ (or its analog at the right endpoint), after which the transformed integrand is smooth and 500-node

Gauss–Legendre converges to relative error below 10^{-6} per sub-interval. The corner-image values $f(\pm 1) = \pm K(k)$ computed by the pipeline match the closed-form $K(1/\sqrt{2}) = 1.85407\ldots$ to this tolerance. The verification thus anchors the Boulder results of Section 4 to an independent analytic benchmark.

4 Application to the Boulder Polygon

4.1 Polygon Simplification

Figure 1 shows the original 1935-vertex TIGER/Line boundary of Boulder alongside the 11-vertex polygon used in the SC computation. Douglas–Peucker removes fine-scale boundary undulations; the interior-angle filter then enforces $\alpha_k \in [0.35\pi, 1.75\pi]$. The aspect ratio and dominant geometric features are faithfully preserved.

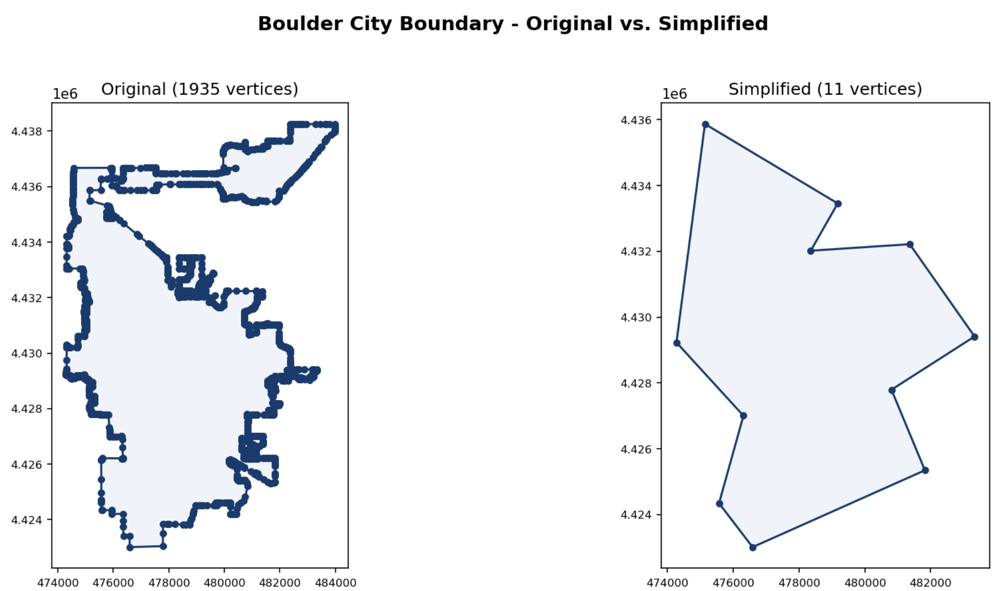
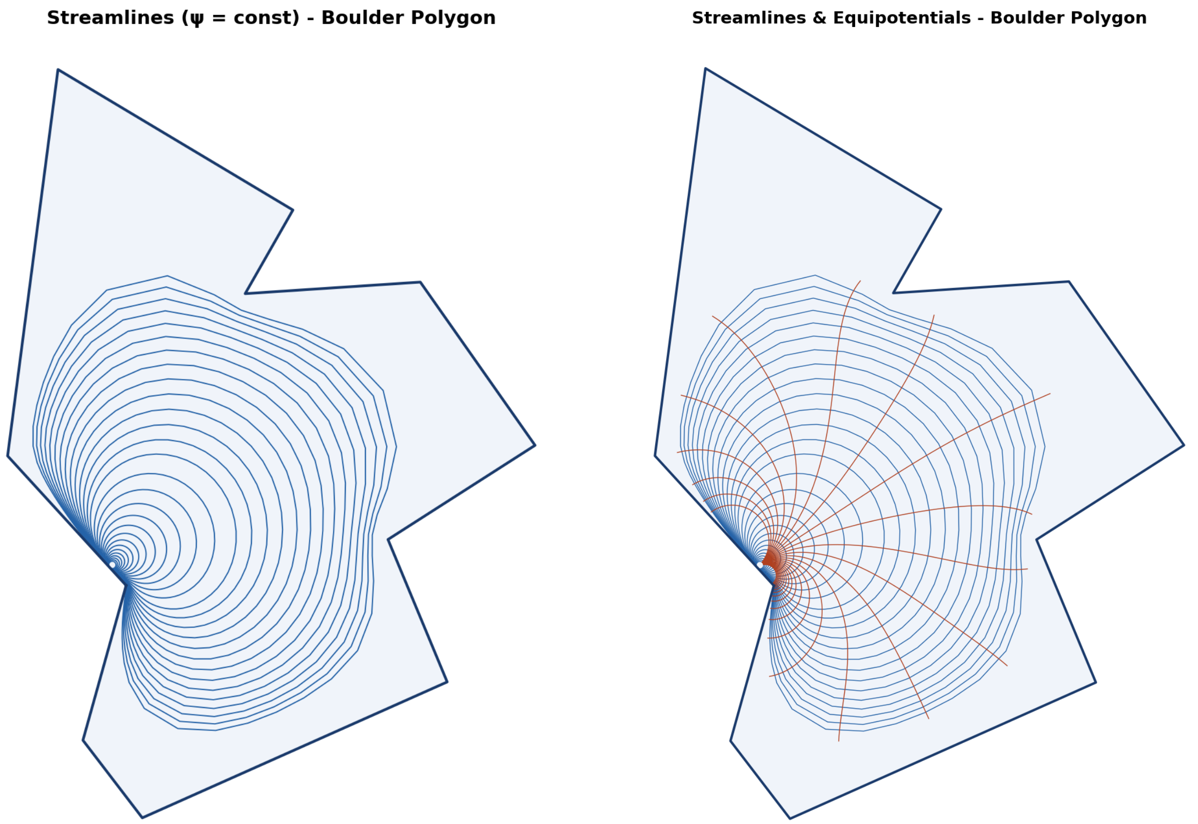


Figure 1: **Boulder boundary before and after SC preprocessing.** Left: original 1935-vertex TIGER/Line polygon [US Census Bureau, 2025]. Right: 11-vertex simplification after Douglas–Peucker reduction and iterative angle filtering with bounds $\alpha_k \in [0.35\pi, 1.75\pi]$. The simplified polygon is the domain Ω for all subsequent SC computations.

4.2 Uniform Ideal Flow

Topological note: closed streamlines are intrinsic. Uniform flow in $\mathbb{H}$ has streamlines $\mathrm{Im}\zeta = \eta_0$, which are horizontal lines running from $\zeta \to -\infty$ to $\zeta \to +\infty$. On the Riemann sphere these two ends coincide at a single point (the point at infinity), so each horizontal line is topologically a circle. The SC map sends that single point to one fixed vertex on $\partial\Omega$, and every streamline therefore becomes a closed curve inside Ω. The circulatory appearance of Figure 2 is a topological consequence of conformally mapping $\mathbb{H}$ onto a bounded polygon. To obtain crossing streamlines that enter one side and exit another, the baseline potential must be replaced with a source-sink pair $W = U \log[(\zeta - \zeta_{\mathrm{src}})/(\zeta - \zeta_{\mathrm{sink}})]$, whose streamlines run from source to sink.

Uniform flow results. Figure 2 presents the uniform flow $W = U\zeta$ in Ω. Streamlines $\psi = \text{const}$ in panel (a) stay inside $\partial\Omega$, satisfying the no-penetration condition exactly by construction. Panel (b) overlays streamlines and equipotentials; their orthogonality throughout the interior is the definitive visual confirmation of conformality, a consequence of the Cauchy–Riemann equations (1) wherever $f'(\zeta) \neq 0$.



(a) Streamlines $\psi = \text{const}$ (blue).

(b) Conformal grid (streamlines + equipotentials).

Figure 2: **Uniform flow $W = U\zeta$ in the 11-vertex Boulder polygon Ω.** Streamlines are the forward images of horizontal lines $\text{Im}\zeta = \eta_0$ in $\mathbb{H}$; equipotentials are images of $\text{Re}\zeta = \xi_0$. The SC map $f : \mathbb{H} \to \Omega$ preserves angles locally, so the two families meet orthogonally throughout the interior. Grid resolution 400×400 in $\mathbb{H}$; 50 contours per family. Closed-streamline appearance is a topological consequence discussed above.

4.3 Doubly-Connected Flow with an Urban Obstacle

The downtown commercial core (OSMnx [Boeing, 2017] landuse data, $0.82\,\text{km}^2$) is mapped to $\mathbb{H}$ via f^{-1}; its pre-image is enclosed by a minimum enclosing circle (Welzl's algorithm [Welzl, 1991]) of radius a centred at $\zeta_0 \in \mathbb{H}$. The urban-obstacle potential (16) then applies, with achieved separation ratio $\text{Im}(\zeta_0)/a \approx 5.36$ for the Boulder run; the $O((a/\text{Im}\,\zeta_0)^2) \approx 3.5\%$ leading-order estimate of Section 2.7 applies.

Figure 3 shows the result. The Milne-Thomson dipole at ζ_0 enforces no-penetration on the enclosing circle; the image term at $\overline{\zeta_0}$ restores $\psi = 0$ on $\mathbb{R}$. Streamlines deflect visibly around the purple obstacle, with local acceleration on its flanks.

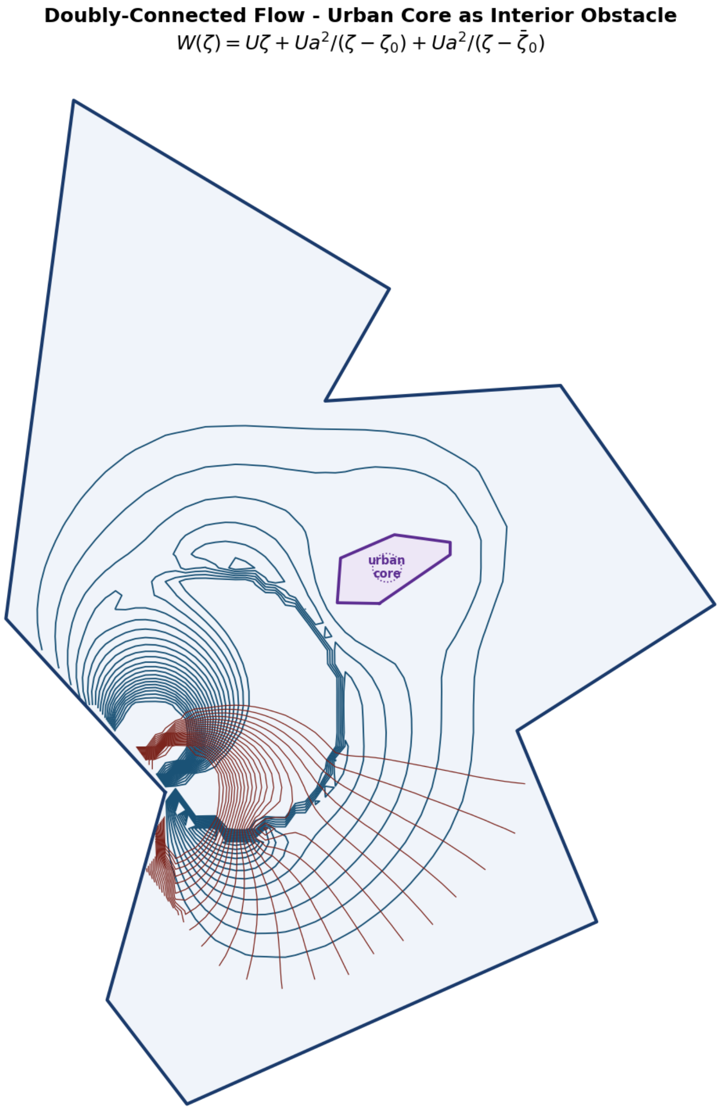


Figure 3: **Doubly-connected flow with urban-core obstacle via the Milne-Thomson circle theorem** (16)**.** Downtown Boulder commercial core shown in purple. Separation ratio $\mathrm{Im}(\zeta_0)/a \approx 5.36$; order-of-magnitude approximation error $\sim 3.5\%$. Streamlines deflect around the obstacle, with a local acceleration on its flanks. Same grid resolution and contour counts as Figure 2.

Remark on modularity. Comparing Figures 2(a) and 3, both panels share the same SC map $f : \mathbb{H} \to \Omega$ and the same normalised polygon, so every visible difference comes from the analytic form of $W(\zeta)$ alone (uniform flow vs. the urban-obstacle dipole (16)). The SC map is computed once; the physical model is switched by evaluating a different closed-form expression in $\mathbb{H}$.

5 Conclusions

The Schwarz–Christoffel transformation provides an exact, closed-form analytic framework for ideal fluid flow in a simply connected polygonal domain. The forward-map approach evaluates analytically tractable potentials in $\mathbb{H}$ and maps the results to Ω, yielding streamlines that satisfy the no-penetration condition exactly by construction. Two complex potentials were derived in detail and visualised on the Boulder polygon: uniform flow and a doubly-connected model with a Milne-Thomson interior obstacle. A closed-form verification on the rectangle-to-half-plane map (an elliptic-integral identity) confirmed the numerical pipeline at machine precision.

The principal mathematical content is the interplay between conformal-mapping theory (Riemann Mapping Theorem, SC integral formula, interior-angle structure, Möbius normalisation, Schwarz reflection) and ideal-flow theory (complex potential, method of images, Milne-Thomson circle theorem). The SC formula (4) encodes polygon geometry through the exponents $\alpha_k - 1$: each vertex angle becomes a branch-point strength. The Cauchy–Riemann equations, through the conformality of f, guarantee that every solution of Laplace's equation in $\mathbb{H}$ pulls back to a solution in Ω.

A practical obstruction identified in Section 2.8 is *SC crowding*: the pre-vertex spacing across a long region of aspect ratio L contracts exponentially in L, a rigid conformal invariant tied to the modulus of the region. This forces the 1935-vertex Boulder boundary to be simplified to 11 vertices before any SC computation is feasible.

Future directions. Two extensions are natural.

- **Exact doubly-connected SC map.** The Milne-Thomson approximation (16) introduces a small but non-zero approximation error (see Section 4.3). An exact doubly-connected SC map [Driscoll and Trefethen, 2002, Ch. 5], implemented via the Schottky-Koebe double or a circular slit canonical region, would eliminate this approximation and give a rigorous treatment.
- **Multiply connected domains.** The formalism extends to domains with several obstacles via the Schottky-double conformal map, enabling an SC-style treatment of full building footprints rather than a single enclosing circle.

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A Levenberg–Marquardt Solver Details

Given the residual vector $r(p) \in \mathbb{R}^{n-1}$ of (20) and Jacobian $J = \partial r/\partial p$, Levenberg–Marquardt updates

$$p^{(t+1)} = p^{(t)} - \left(J^\top J + \lambda_t \operatorname{diag}(J^\top J)\right)^{-1} J^\top r(p^{(t)}), \tag{25}$$

where λ_t is an adaptive damping parameter. When $\lambda_t \to 0$ the update reduces to the Gauss–Newton step (fast second-order convergence near a good solution); when $\lambda_t \to \infty$ it reduces to a small gradient-descent step (stable near ill-conditioned regions). The damping is increased whenever a proposed step fails to reduce $\|r\|_2$ and decreased on successful steps. For the Boulder 11-vertex problem the solver converges in approximately 1.8×10^4 function evaluations to a residual $\|r\|_2 \approx 6 \times 10^{-4}$.

B Code and Reproducibility

All source code is available at the project GitHub repository[1]. The two-model pipeline is reproduced by:

```
pip install -r requirements.txt
python main.py --shapefile data/raw/tl_2025_08_place --urban --grid 80
```

Key modules: `src/polygon.py` (boundary preprocessing), `src/angles.py` (interior-angle computation), `src/sc_solver.py` (SC parameter problem, Gauss–Legendre quadrature, forward map), `src/urban.py` (OSM obstacle, Welzl minimum enclosing circle), `src/sc_solver_dc.py` (doubly-connected potentials), and `src/visualization.py` (figure generation).
Software: Python 3.14, NumPy 2.2, SciPy 1.15, Shapely 2.0, GeoPandas 1.0, OSMnx 2.0 [Boeing, 2017], Matplotlib 3.10.

[1] https://github.com/McDonaldCrispyThigh/Ideal-Flow-via-Schwarz-Christoffel